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Jeans-Independent Holdback Bypass for AMR Refinement in Cosmological Simulations

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ABSTRACT

Cosmological AMR simulations employ a *holdback* mechanism that progressively releases refinement levels as the universe expands, preventing premature over-refinement at high redshift. However, this mechanism blocks *all* refinement criteria simultaneously, including the Jeans length criterion, which is a numerical stability requirement rather than merely a resolution preference. When a holdback level is finally released, a large number of cells suddenly satisfy the refinement condition, triggering an abrupt starburst. We implement a `jeans_bypass_holdback` option that allows the Jeans criterion to operate independently of the holdback, with a configurable level limit (`n_jeans_bypass_levels`) to prevent uncontrolled refinement cascades. We discuss the interplay between the Jeans criterion, the polytropic equation of state, and the density threshold `d_jeans_thre`, showing that the polytropic temperature floor provides a natural self-regulation mechanism for the bypass.

Keywords: methods: numerical — cosmology: simulations — stars: formation — hydrodynamics

1. AMR REFINEMENT IN RAMSES

In RAMSES, each cell at level $\ell < N_{\rm levelmax}$ is tested for refinement at every coarse time-step. The decision to subdivide a cell into $2^{N_{\rm dim}}$ children is governed by a combination of physical, geometric, and cosmological criteria.

1.1. *Refinement Logic*

The overall refinement flag for a cell is determined by the following Boolean expression:

$$\mathcal{R} = \neg\,\mathcal{H} \;\wedge\; \mathcal{G} \;\wedge\; \big(\mathcal{M} \vee \mathcal{S} \vee \mathcal{D} \vee \mathcal{P} \vee \mathcal{U} \vee \mathcal{J}\big), \qquad (1)$$

where each symbol is defined below.

1.1.1. *Physical Criteria (OR)*

1. **Quasi-Lagrangian mass criterion ($\mathcal{M}$).** The cell is flagged if its gas density exceeds a level-dependent threshold:

$$\rho_{\rm cell} \;\geq\; m_{\rm refine}(\ell)\,\frac{m_{\rm sph}}{V_{\rm cell}}, \qquad (2)$$

where $m_{\rm sph}$ is the effective SPH kernel mass and $V_{\rm cell} = (\Delta x_\ell)^3$. This criterion ensures that regions containing roughly $m_{\rm refine}$ or more particles per cell are refined, providing a quasi-Lagrangian resolution that tracks gravitational collapse. Controlled by the namelist array `m_refine(1:nlevelmax)`.

2. **Sink particle forcing ($\mathcal{S}$).** If `sink_refine=.true.`, any cell containing a sink cloud particle (identified by `idp` < 0 and `tp` $= 0$) is unconditionally flagged up to $N_{\rm levelmax}$, guaranteeing maximum resolution around each accretion region.

3. **Density gradient ($\mathcal{D}$).** The normalised density gradient between a cell and each of its $2N_{\rm dim}$ face-neighbours is computed as

$$\epsilon_\rho = 2\,\max\!\left(\frac{|\rho_+ - \rho_0|}{|\rho_+ + \rho_0| + \rho_{\rm floor}},\; \frac{|\rho_0 - \rho_-|}{|\rho_0 + \rho_-| + \rho_{\rm floor}}\right). \qquad (3)$$

If $\epsilon_\rho >$ `err_grad_d` along any axis, the cell is flagged.

4. **Pressure gradient ($\mathcal{P}$).** Analogous to the density gradient but applied to the thermal pressure:

$$\epsilon_P = 2\,\max\!\left(\frac{|P_+ - P_0|}{|P_+ + P_0| + P_{\rm floor}},\; \frac{|P_0 - P_-|}{|P_0 + P_-| + P_{\rm floor}}\right), \qquad (4)$$

flagging when $\epsilon_P >$ `err_grad_p`.

5. **Velocity gradient ($\mathcal{U}$).** A normalised velocity jump relative to the local sound speed:

$$\epsilon_v = 2\,\max\!\left(\frac{|v_+ - v_0|}{c_{s,+} + c_{s,0} + |v_+| + |v_0| + v_{\rm floor}},\; \frac{|v_0 -}{c_{s,0} + c_{s,-} + |v_0}\right. \qquad (5)$$

flagging when $\epsilon_v >$ `err_grad_u` in any component.

6. **Jeans length ($\mathcal{J}$).** When `jeans_refine(`ℓ`)` $= N_J > 0$, the cell is flagged if the local Jeans length is shorter than N_J cell widths:

$$\lambda_{\rm J} = \sqrt{\frac{\pi\, c_s^2}{G\,\rho}} < N_J\,\Delta x_\ell, \tag{6}$$

implementing the Truelove et al. (1997) condition to prevent artificial fragmentation in self-gravitating gas. An additional density threshold `d_jeans_thre` restricts this criterion to cells with $n_{\rm H} \geq$ `d_jeans_thre`.

1.1.2. *Geometry Mask (AND)*

Ellipsoidal region ($\mathcal{G}$). When `r_refine(`ℓ`)` > -1, only cells inside a generalised ellipsoidal region are permitted to refine. The test is

$$\left(\frac{2|\Delta x|}{r}\right)^{e_r} + \left(\frac{2|\Delta y|}{a\,r}\right)^{e_r} + \left(\frac{2|\Delta z|}{b\,r}\right)^{e_r} < 1, \tag{7}$$

where $r =$ `r_refine`, $a =$ `a_refine`, $b =$ `b_refine`, and $e_r =$ `exp_refine`. Cells outside this region are *unflagged* regardless of physical criteria, restricting high resolution to a zoom-in target.

1.1.3. *Cosmological Holdback (NOT)*

Resolution holdback ($\mathcal{H}$). In cosmological runs with cooling, overly eager refinement at high redshift can exhaust memory before the epoch of interest. When `q_refine_holdback=.true.` and the physical cell size $\Delta x_\ell < 4^{1/N_{\rm dim}}\,\Delta x_{N_{\rm levelmax}}/a_{\rm exp}$ for levels $\ell >$ `nlevelmax_part`$+3$, the *entire* level is blocked from further refinement:

$$\mathcal{H}(\ell) = \big[\ell > \ell_{\rm part}{+}\ell_{\rm collapse}\big] \wedge \big[\Delta x_\ell < 4^{1/N_{\rm dim}}\,\Delta x_{\rm min}/a\big], \tag{8}$$

where $\ell_{\rm part}$ is the particle initial-condition level (typically $\ell_{\rm min}$) and $\ell_{\rm collapse} = 3$. The first holdback level $\ell_{\rm hb}$ is the smallest ℓ satisfying Equation (8):

$$\ell_{\rm hb} = \min\big\{\ell > \ell_{\rm part}{+}\ell_{\rm collapse}\ :\ \Delta x_\ell < 4^{1/N_{\rm dim}}\,\Delta x_{\rm min}/a\big\}. \tag{9}$$

1.2. *Buffer Zones*

After applying Equation (1), flagged cells are expanded by `nexpand(`ℓ`)` cell widths in each direction via a dilation operator (`smooth_fine`). This cubic buffer ensures smooth refinement boundaries, preventing the Godunov solver from encountering abrupt level transitions.

2. THE STARBURST PROBLEM

The holdback mechanism blocks *all* refinement criteria at levels $\ell \geq \ell_{\rm hb}$, including the Jeans criterion. This creates two related problems:

Artificial fragmentation.—The Jeans length criterion $\mathcal{J}$ differs fundamentally from the other refinement triggers: it is a *numerical stability* requirement (Truelove et al. 1997), not merely a resolution preference. Violating the Jeans condition leads to artificial fragmentation — a qualitatively wrong result — regardless of epoch. Yet in the standard RAMSES logic (Equation 1), the cosmological holdback $\mathcal{H}$ blocks $\mathcal{J}$ uniformly with all other criteria. At high redshift, dense self-gravitating regions may therefore be held at insufficient resolution, producing spurious fragmentation that propagates into lower-redshift structure.

Abrupt SFR transition.—When the holdback releases level ℓ (i.e. $\mathcal{H}(\ell)$ transitions from true to false), all accumulated criteria fire simultaneously. Many cells that have been gravitationally collapsing during the holdback period now suddenly satisfy the refinement condition, leading to a burst of new cells and an abrupt spike in the star formation rate.

3. JEANS BYPASS: DESIGN AND IMPLEMENTATION

3.1. *Modified Refinement Logic*

The key observation is that the Jeans length criterion is *cell-local*: it depends only on the density and temperature of the cell itself (`uold`), with no neighbor information required. This allows a lightweight evaluation even when the full refinement machinery (which requires neighbor gathering for $\mathcal{D}$, $\mathcal{P}$, $\mathcal{U}$) is skipped by the holdback.

With `jeans_bypass_holdback=.true.`, the refinement logic becomes

$$\mathcal{R}' = \Big[\neg\,\mathcal{H} \wedge \mathcal{G} \wedge \big(\mathcal{M}{\vee}\mathcal{S}{\vee}\mathcal{D}{\vee}\mathcal{P}{\vee}\mathcal{U}{\vee}\mathcal{J}\big)\Big] \vee \Big[\mathcal{H} \wedge \mathcal{J} \wedge (\ell < \ell_{\rm hb}{+}n_{\rm byp} \tag{10}$$

where $n_{\rm bypass} \equiv$ `n_jeans_bypass_levels` (default 1). The geometry mask $\mathcal{G}$ still applies through the non-holdback branch, preserving zoom-in spatial restrictions. Only the temporal holdback $\mathcal{H}$ is bypassed for $\mathcal{J}$, and only within a limited number of levels.

This modification is analogous to the existing treatment of sink particles: $\mathcal{S}$ is already evaluated inside the holdback-protected block, but unlike $\mathcal{J}$ it only triggers in cells that already host a sink. Extending the same exemption to $\mathcal{J}$ ensures numerical integrity for gas that is *about to* form a sink.

3.2. *Level Limitation: Preventing Cascades*

Without a level limit ($n_{\rm bypass} \to \infty$), the Jeans bypass creates a catastrophic cascade. When a cell at the holdback level $\ell_{\rm hb}$ is flagged by the Jeans criterion, child

cells are created at level $\ell_{\rm hb}+1$. Since level $\ell_{\rm hb}+1$ is also in the holdback zone, the bypass applies again, flagging cells at $\ell_{\rm hb}+2$, and so on until $\ell_{\rm max}$. This rapidly exhausts available memory.

The cascade occurs because the Jeans length in radiatively cooling gas (below the star-formation density threshold) shrinks without bound as the temperature drops. There is no physical mechanism to halt the Jeans criterion from triggering at every successive level — only the polytropic temperature floor (Section 4) or a hard level limit can stop it.

Table 1 shows the progression observed in a test run without level limit.

The memory footprint grew from 81% to 88.5% in a single coarse step before crashing. The fine-step timestep also decreased by 8× due to subcycling at levels 15–17.

With $n_{\rm bypass}=1$, only level $\ell_{\rm hb}=14$ is bypassed, creating cells at level 15 but no further. This provides a controlled pre-refinement before the holdback releases.

4. ROLE OF THE POLYTROPIC TEMPERATURE FLOOR

4.1. *Jeans Length in the Polytropic Regime*

In RAMSES, the star-formation module enforces a polytropic equation of state above the density threshold $n_\star$:

$$T(n) = T_{2,\star}\left(\frac{n}{n_\star}\right)^{g_\star-1}, \qquad n \geq n_\star, \tag{11}$$

with default parameters $T_{2,\star}=750\,{\rm K}$, $n_\star=0.1\,{\rm cm}^{-3}$, $g_\star=5/3$. This floor is applied in the cooling routine and is already reflected in `uold`, so the Jeans length computation in `jeans_length_refine()` automatically includes the polytropic stabilisation.

Substituting Equation (11) into Equation (6):

$$\lambda_{\rm J} \propto \sqrt{\frac{T}{\rho}} \propto \sqrt{\rho^{g_\star-1}/\rho} = \rho^{(g_\star-2)/2}. \tag{12}$$

For $g_\star=5/3$: $\lambda_{\rm J}\propto\rho^{-1/6}$. The Jeans length *decreases very slowly* with density — one or two additional levels of refinement are typically sufficient to resolve it.

In contrast, for gas below $n_\star$ (no polytropic floor), radiative cooling can drive T to very low values while ρ remains moderate. This yields $\lambda_{\rm J}\ll\Delta x_\ell$ at many levels, causing the cascade.

4.2. *Self-regulation via `d_jeans_thre`*

The density threshold parameter `d_jeans_thre` (in units of $n_{\rm H}\,[{\rm cm}^{-3}]$) restricts the Jeans criterion to cells above a given density:

$$\mathcal{J}({\rm cell}) = \begin{cases} N_J\,\Delta x_\ell \geq \lambda_{\rm J} & \text{if } n \geq \texttt{d_jeans_thre}, \\ \text{false} & \text{otherwise.} \end{cases} \tag{13}$$

Setting `d_jeans_thre` $\approx n_\star$ restricts the Jeans criterion to the polytropic regime. The cascade then self-limits through the following mechanism:

1. At level $\ell_{\rm hb}$: only cells with $n\geq n_\star$ are checked. The polytropic floor guarantees $T\geq T_{2,\star}$, so $\lambda_{\rm J}$ is relatively large.

2. If $N_J\,\Delta x_{\ell_{\rm hb}} > \lambda_{\rm J}$, the cell is flagged and children at $\ell_{\rm hb}+1$ are created.

3. At level $\ell_{\rm hb}+1$: Δx is halved, density is approximately unchanged, $\lambda_{\rm J}$ is approximately unchanged.

4. $N_J\,\Delta x_{\ell_{\rm hb}+1} = N_J\,\Delta x_{\ell_{\rm hb}}/2 < \lambda_{\rm J} \Rightarrow$ Jeans criterion is *satisfied* $\Rightarrow$ no further flagging $\Rightarrow$ cascade stops naturally.

This provides a *physics-based* self-regulation, complementary to the *hard limit* imposed by `n_jeans_bypass_levels`.

When `d_jeans_thre` $=0$ (default), *all* cells are checked, including low-density gas where cooling produces arbitrarily small $\lambda_{\rm J}$. This is the root cause of the unlimited cascade observed in Section 3.2.

5. IMPLEMENTATION DETAILS

5.1. *Code Structure*

The implementation modifies two source files (plus parameter declaration and namelist reading).

`flag_utils.kjhan.f90` — `userflag_fine()`—The holdback early-return is modified from a simple `return` to a conditional block:

```
if(prevent_refine .and. q_refine_holdback) then
   if(jeans_bypass_holdback .and. hydro &
      .and. n_jeans_bypass_levels > 0) then
      ! Find the first holdback level
      dx_min = (0.5D0**nlevelmax)*scale
      holdback_threshold = &
        (4d0**(1d0/ndim))*(dx_min/aexp)
      holdback_first = nlevelmax + 1
      do jlevel = nlevelmax_part &
         + nlevel_collapse+1, nlevelmax
         dx_test = (0.5d0**jlevel)*scale
         if(dx_test < holdback_threshold) then
            holdback_first = jlevel
            exit
         endif
      end do
      ! Only bypass within allowed range
      if(ilevel < holdback_first &
         + n_jeans_bypass_levels) then
         call jeans_only_flag(ilevel)
      endif
   endif
   return
endif
```

Table 1. Cascade refinement without level limit ($a = 0.164$, $\ell_{\rm hb} = 14$, $\ell_{\rm max} = 17$). All levels ≥ 14 are in the holdback zone.

Level	$\Delta x/\Delta x_{\rm min}$	Holdback	Bypass	$\mathcal{J}$ fires	Result
≤ 12	≥ 32	No	—	Yes (normal)	Normal
13	16	No	—	Yes (normal)	Normal
14	8	Yes	Active	Yes	$\rightarrow$ lv. 15 created
15	4	Yes	Active	Yes	$\rightarrow$ lv. 16 created
16	2	Yes	Active	Yes	$\rightarrow$ lv. 17 created
17	1	$\ell_{\rm max}$	—	—	OOM crash

Table 2. Two complementary mechanisms for preventing the bypass cascade.

Mechanism	Type	Behaviour
`n_jeans_bypass_levels`	Hard limit	Caps extra levels regardless of physics
`d_jeans_thre` $\approx n_\star$	Physics-based	Polytropic floor limits $\lambda_{\rm J}$ decrease

hydro_flag.kjhan.f90 — *jeans_only_flag()*—A new lightweight subroutine that:

1. Loops over active grids with OpenMP parallelism.
2. Computes cell indices without neighbor gathering.
3. Calls the existing `jeans_length_refine()` subroutine (no code duplication).
4. Sets `flag1` for flagged cells and updates `nflag`.

This is possible because `jeans_length_refine()` only accesses cell-local data (`uold(ind_cell,...)`), unlike the gradient-based criteria $\mathcal{D}$, $\mathcal{P}$, $\mathcal{U}$.

5.2. *Safety Properties*

- **Default off**: With `jeans_bypass_holdback=.false.` (default), the new code path is never entered. Bit-identical to the unmodified code.
- **No neighbor access**: `jeans_only_flag()` is safe to call at any point in the refinement cycle.
- **OMP safety**: `flag1` writes are grid-partitioned; `nflag` uses reduction.
- **Hard level limit**: `n_jeans_bypass_levels` provides a fail-safe even if `d_jeans_thre` is misconfigured.

6. SUMMARY OF PARAMETERS

Table 3 lists all refinement-related namelist parameters, including the new bypass parameters.

The recommended configuration for cosmological runs with star formation is:

```
&REFINE_PARAMS
m_refine           = 9*8.,
jeans_refine       = 9*4.,
jeans_bypass_holdback  = .true.
n_jeans_bypass_levels  = 1
d_jeans_thre       = 0.1      ! = n_star
q_refine_holdback = .true.
/
```

7. TEST RESULTS

All tests use the HR5 medium-box setup: $\ell_{\rm min} = 9$, $\ell_{\rm max} = 17$, 12 MPI ranks, OMP=5, restarting from output 5 ($a = 0.164$, $z \approx 5.1$, nstep_coarse=241).

7.1. *Bit-identical Verification*

With `jeans_bypass_holdback=.false.` (default) and no `jeans_refine`, the new binary produces results identical to the reference run: econs=6.13×10^{-4} at step 242, econs=6.71×10^{-4} at step 243.

7.2. *Unlimited Bypass*

With `jeans_refine=9*4.`, `jeans_bypass_holdback=.true.`, `d_jeans_thre=0`, and no level limit: cascade refinement from level 14 through 17 within a single coarse step. Memory grew from 81% to 88.5% before OOM crash. Fine-step timestep decreased by 8× due to subcycling.

Table 3. AMR refinement parameters in the `&AMR_PARAMS` and `&REFINE_PARAMS` namelists. New parameters are marked with ⋆.

Parameter	Default	Role
`m_refine`	8.0	Lagrangian mass threshold (per level)
`sink_refine`	F	Force max level around sinks
`err_grad_d`	−1	Density gradient threshold
`err_grad_p`	−1	Pressure gradient threshold
`err_grad_u`	−1	Velocity gradient threshold
`jeans_refine`	−1	Jeans length in cells (per level)
`d_jeans_thre`	0	Density threshold for Jeans ($n_{\rm H}\,{\rm cm}^{-3}$)
`r_refine`	−1	Geometry region diameter (per level)
`a_refine`	1.0	Ellipticity Y/X
`b_refine`	1.0	Ellipticity Z/X
`exp_refine`	2.0	Norm exponent
`x/y/z_refine`	0.5	Region centre
`nexpand`	1	Buffer zone width (per level)
`q_refine_holdback`	F	Cosmological holdback
⋆ `jeans_bypass_holdback`	F	Enable Jeans bypass of holdback
⋆ `n_jeans_bypass_levels`	1	Max extra levels beyond $\ell_{\rm hb}$

7.3. *Limited Bypass ($n_{\rm bypass} = 1$)*

With `n_jeans_bypass_levels=1`: only level 14 is bypassed, cells at level 15 may be created but no further cascade.

7.4. *Future: `d_jeans_thre` $= n_\star$*

A test with `d_jeans_thre=0.1` will verify the physics-based self-regulation, potentially allowing $n_{\rm bypass} > 1$ without cascade, since only gas in the polytropic regime would trigger the Jeans criterion.

8. COOLING MODULE COMPARISON: RAMSES VS GRACKLE

The cooling module is a critical component of the simulation, as it determines the equilibrium temperature of the gas and hence the Jeans length. Different cooling implementations can therefore affect the refinement behaviour through the Jeans criterion $\mathcal{J}$.

We compare the standard RAMSES cooling module (implicit Newton–Raphson solver with Courty UV background and Cloudy 07 metal cooling tables) against the Grackle equilibrium cooling library (Smith et al. 2017) using the Townsend (2009) exact Y-function integration with Haardt & Madau (2012) UV background.

8.1. *Comparison on the Density–Temperature Plane*

To quantify the systematic differences between the two cooling models, we compute the net cooling rate $\Lambda_{\rm net}(n_{\rm H}, T)$ on a grid of 60×256 points spanning $n_{\rm H} = 10^{-8}$–$10^{4}\,{\rm cm}^{-3}$ and $T = 10$–$10^{9}\,{\rm K}$ at $z = 5.1$ (the redshift of the test restart).

Low metallicity ($Z = 10^{-3}\,Z_\odot$).—Figure 1 shows the comparison at the simulation average metallicity (`z_ave` $= 10^{-3}$). At this low metallicity, the dominant cooling processes are H/He line emission and bremsstrahlung, while the dominant heating is UV photoheating.

The lower-left panel shows the log ratio of the cooling rates: $\log_{10}(|\Lambda_{\rm RAMSES}|/|\Lambda_{\rm Grackle}|)$. Contour lines mark the equilibrium temperature ($\Lambda_{\rm net} = 0$) for each model: the yellow solid line for RAMSES and the green dashed line for Grackle. The two equilibrium curves differ most at low densities ($n_{\rm H} < 1\,{\rm cm}^{-3}$), reflecting the different UV backgrounds (Courty vs HM2012).

The lower-right panel shows the mean molecular weight difference $\Delta\mu = \mu_{\rm RAMSES} - \mu_{\rm Grackle}$. At $T > 10^5\,{\rm K}$, the RAMSES CIE calculation gives a higher μ than the Grackle non-LTE treatment, while at intermediate temperatures (10^4–$10^5\,{\rm K}$), the density-dependent Grackle μ shows significant departures from the RAMSES density-independent equilibrium.

Solar metallicity ($Z \approx Z_\odot$).—Figure 2 shows the same comparison at solar metallicity, where metal line cooling dominates the temperature range 10^4–$10^{5.5}\,{\rm K}$. The differences between the two models are more pronounced here: the Grackle table includes a richer set of metal species and density-dependent ionisation corrections, leading to up to $\sim 10\times$ stronger cooling in the $10^{4.5}$–

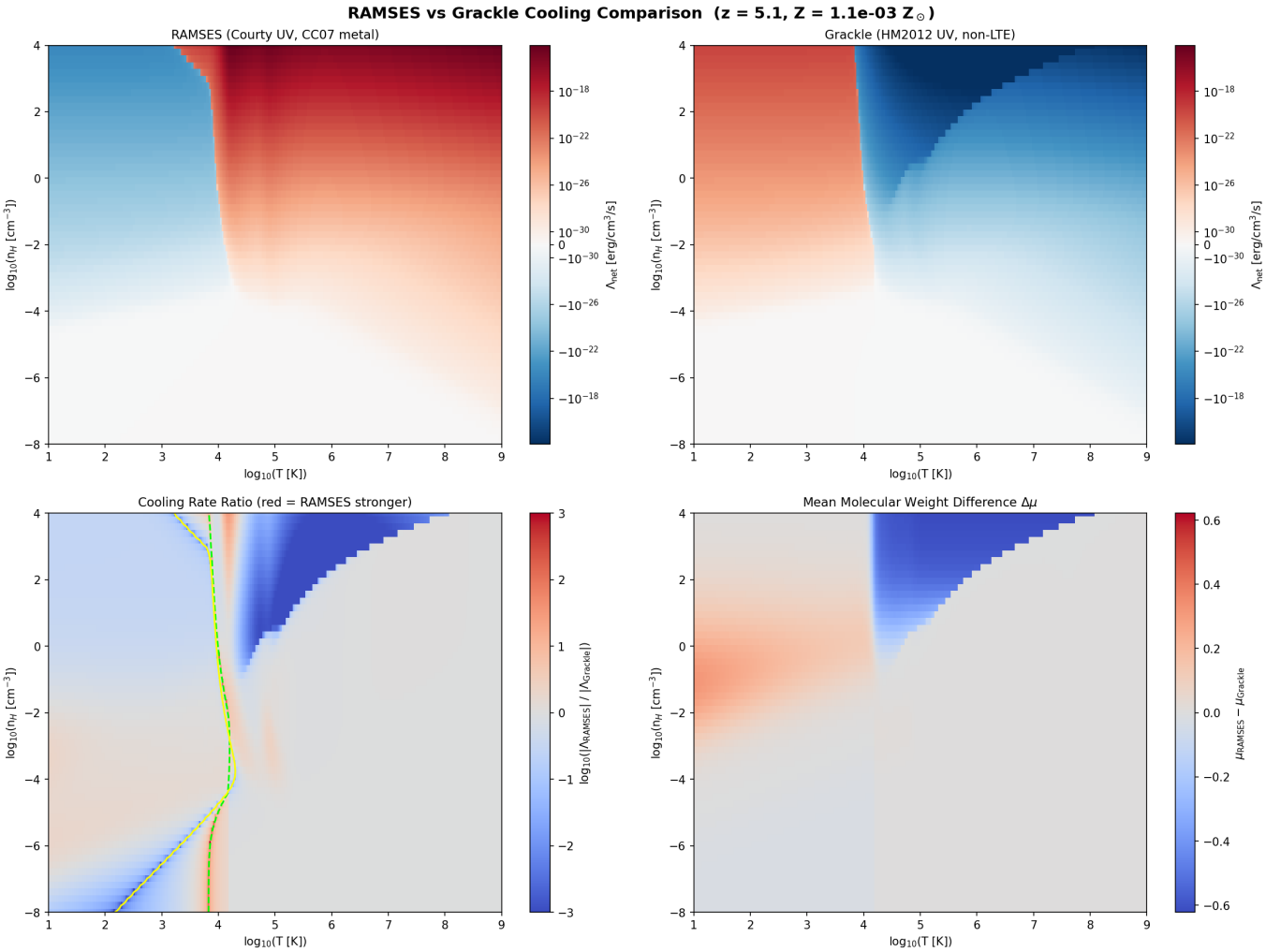

Figure 1. Comparison of the RAMSES and Grackle net cooling rates at $z = 5.1$ and $Z = 1.1 \times 10^{-3}\, Z_\odot$. *Top left:* RAMSES net cooling rate (Courty UV, CC07 metals). *Top right:* Grackle net cooling rate (HM2012 UV, non-LTE). Red indicates cooling (energy loss), blue indicates heating (energy gain). *Bottom left:* Log ratio of the absolute cooling rates; yellow solid = RAMSES equilibrium ($\Lambda = 0$), green dashed = Grackle equilibrium. *Bottom right:* Mean molecular weight difference $\mu_{\rm RAMSES} - \mu_{\rm Grackle}$.

10^5 K range compared to the CC07 CIE table used by RAMSES.

The equilibrium temperature contours show a larger separation at low densities, where the UV suppression of metal cooling differs between the two models.

8.2. *Implications for the Jeans Criterion*

The key differences between the two cooling models are:

1. **UV background:** The Courty and HM2012 models predict different equilibrium temperatures at low densities, affecting the Jeans length by a factor of ~ 2–3 in the diffuse IGM.

2. **Metal cooling:** At solar metallicity, the Grackle non-LTE metal cooling is significantly stronger than CC07 CIE in the 10^4–10^5 K range, potentially leading to faster cooling and earlier star formation.

3. **Mean molecular weight:** The $\Delta\mu$ of up to ± 0.6 translates directly into a temperature mapping error when converting between T/μ (RAMSES internal variable) and physical temperature T.

These differences motivate the implementation of the Eunha exact cooling integrator (Section 3) as an alternative to the standard RAMSES solver, particularly for high-resolution cosmological simulations where the Jeans criterion is critical for star formation.

9. DISCUSSION

9.1. *Physical Motivation*

The holdback mechanism was designed to prevent over-refinement from the mass criterion at early times.

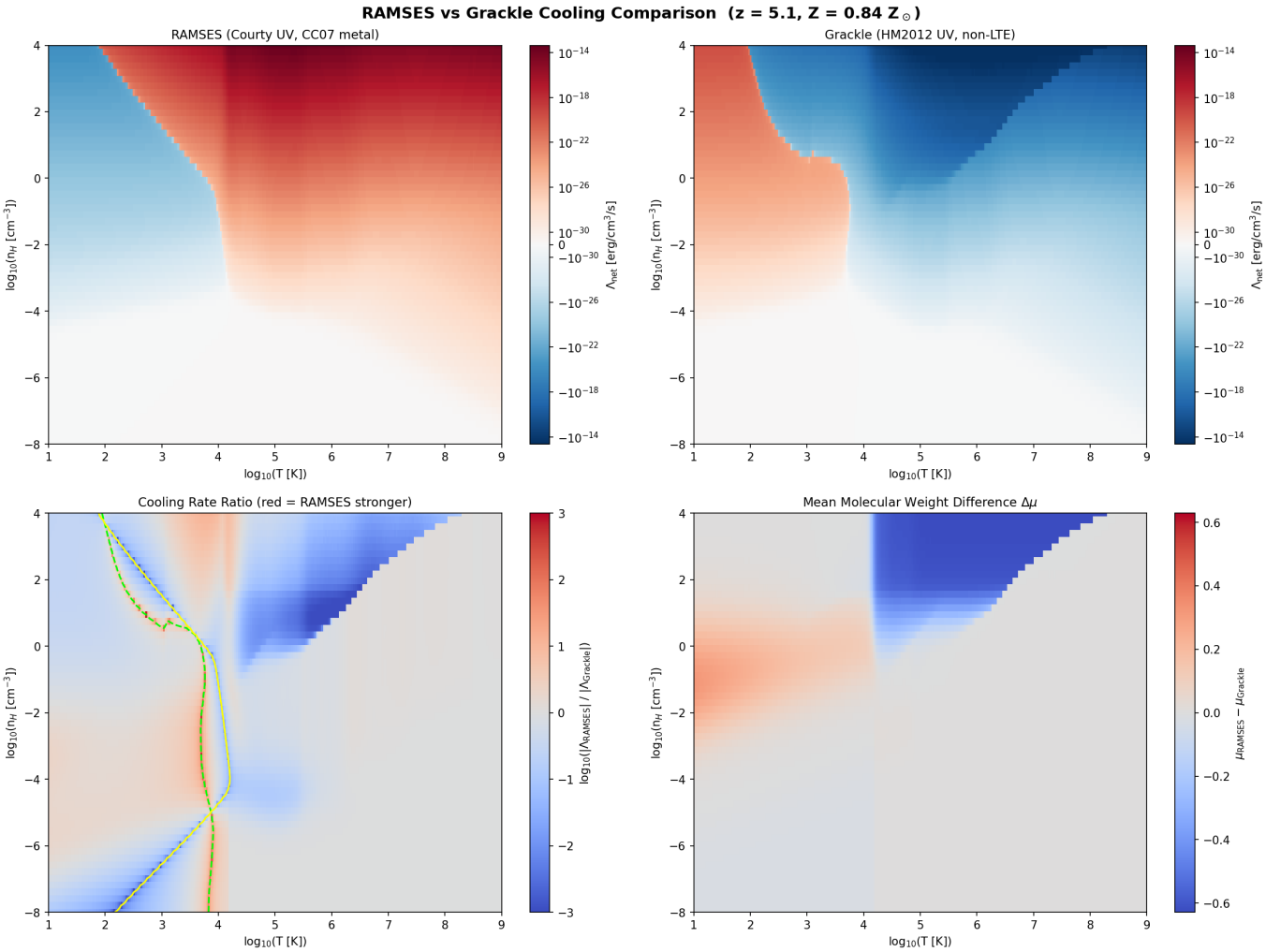

Figure 2. Same as Figure 1 but at solar metallicity ($Z = 0.84\,Z_\odot$). Metal cooling dominates the net rate at $T \sim 10^4$–$10^{5.5}$ K, amplifying the differences between the two models.

However, it inadvertently blocks the Jeans criterion, which has a different physical motivation: resolving the Jeans length prevents artificial fragmentation and ensures numerically converged star formation.

By allowing the Jeans criterion to operate ahead of the holdback release, we expect:

- Dense gas that will form stars is pre-refined before the holdback releases.
- When the holdback releases, the transition is smoother because the densest regions are already at the appropriate resolution.
- The SFR curve shows a gradual onset rather than an abrupt step.

9.2. *Interplay of the Three Parameters*

The bypass is controlled by three parameters that form a layered defence:

Layer 1: `jeans_bypass_holdback` *(on/off)*—Master switch. When off (default), the system is identical to standard RAMSES.

Layer 2: `d_jeans_thre` *(physics-based regulation)*—Restricts the Jeans check to dense gas where the polytropic floor operates. In this regime, $\lambda_J \propto \rho^{-1/6}$ varies slowly, and 1–2 extra refinement levels suffice to resolve it. Provides self-consistent, physics-driven cascade termination.

Layer 3: `n_jeans_bypass_levels` *(hard safety limit)*—Absolute cap on the number of extra levels. Prevents OOM regardless of physics configuration. Essential when `d_jeans_thre=0` (all cells checked).

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